Topic: Memory Segmentation in 8086

Q. Explain the concept of Memory Segmentation in the 8086 microprocessor. How is a 20-bit physical address calculated using a 16-bit segment register and a 16-bit offset?

> **Definition to Remember**

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Memory Segmentation in 8086

The CPU manages these memory blocks using four specialized 16-bit Segment Registers:

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2. Logical vs Physical Address

- **Logical Address:** Format used by programmers `(Segment : Offset)`. e.g., `1000H:0002H`.

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3. Calculating the 20-bit Physical Address

The Bus Interface Unit (BIU) performs this calculation:

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Calculation Example:

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1. **Shift Segment by 10H:** `2000H  10H` = `20000H` (20-bit Base)

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4. Advantages of Segmentation

- **Code Relocation:** Programs can be moved anywhere in memory by changing only the Segment Register; internal offset addresses remain unchanged.

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> **Must-Write Points (for 10 marks)**

> 1. 80

> **Quick Recall**

> `Seg
Offset